When Fusion Confidence Stops Predicting Correction Value: An Operational Study of Audio–Visual Switching

Anonymous research manuscript

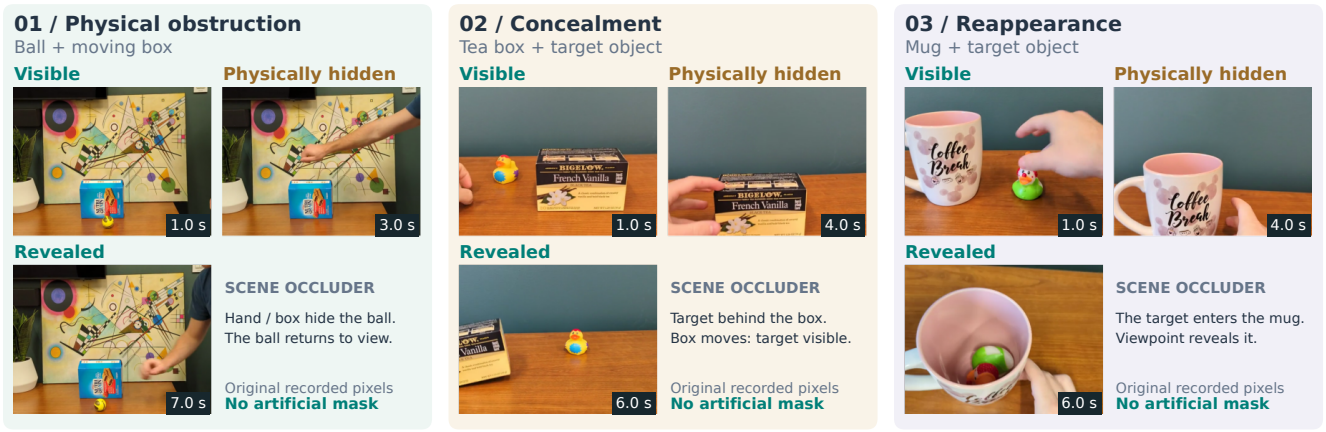


Figure 1. **Occlusion caused by the scene itself.** Three recorded sequences show a ball hidden by a hand and box, a target concealed behind a tea box, and a target hidden inside a mug before a viewpoint change reveals it. All frames are unmodified apart from resizing; timestamps identify the source frames. These companion object-memory demonstrations motivate missing visual evidence; they are separate from the audio–visual routing experiments reported in this paper.

Abstract

An audio–visual selector can confidently replace a correct audio prediction with an incorrect fusion prediction when the visual input changes. We study when this correction confidence ceases to reflect decision value, using controlled input perturbations, actor-separated fitting, and a separately locked external evaluation. On an inspected CREMA-D test, a clean-tuned fixed threshold switches on 53.4% of visually occluded clips and loses 22.93 accuracy points relative to audio; evaluating the same score under a 20% batch quota reports a smaller 7.32-point loss. A calibration audit traces twelve near-zero rescue estimates to an isotonic plateau supported by six non-neutral calibration clips. On 576 clean RAVDESS clips from 24 actors, a reference-distribution support gate rejects every input and trails a margin gate by 4.17 points, with a paired 95% actor interval of $[-6.42, -1.91]$. *Exhaustive calibration-actor subsets further show that lower score error can coincide with less routing benefit, although the effect varies by checkpoint. New exploratory inference across three fusion heads and five checkpoints yields negative static-occluder direct-routing gain in all fifteen fixed-threshold cases and fourteen equal-volume cases. The resulting evaluation principle is to measure actual switching, rescue, harm, and retained opportunities together under the intended deployment rule. We release predictions, executable audits, and a corrected fixed-region certificate whose valid but vacuous bounds expose the available calibration support.*

1. Introduction

Combining sound and video can correct an error that either modality makes alone. This complementarity motivates a second decision: after both experts have executed, should their fused answer replace the audio answer? A selector trained to estimate the value of that replacement appears to make fusion conditional on useful evidence. Yet confidence in the replacement can survive a change that makes the replacement harmful. Figure 1 shows physical occlusion in a separate object-memory demonstration; our routing study quantifies constructed visual stresses, not these natural-scene cases.

The evaluation rule matters. A batch selector can replace only the highest-scoring fraction of a collection, whereas a deployed fixed threshold admits every example above that

threshold. If scores rise under a visual perturbation, the threshold can admit many more replacements than the clean tuning budget suggests. A batch quota hides this increase by construction. Neither a positive score nor a small calibration error establishes that the admitted replacements are beneficial under the changed input distribution.

We investigate this distinction in audio–visual intended-emotion classification. Our aim is an operational reliability analysis, rather than a new successful fusion architecture. We separate an inspected CREMA-D diagnosis from an external RAVDESS evaluation whose procedure was fixed before inference. Within this scope, three mechanisms are measurable: changing switch volume, deteriorating correction quality among selected clips, and limited support for a fitted calibration map. We test practical remedies using common observations and explicit deployment rules. Their failure is part of the result: support rejection learned from one domain can eliminate every useful clean fusion opportunity in another.

The evidence also limits broader interpretations. Both datasets contain performed expressions, and the main visual stresses are constructed. Additional feature, head, and calibration-subset analyses reuse inspected data. They test sensitivity of the reported mechanism, not a newly confirmed method. The central question is therefore precise: *when does confidence in a fusion correction cease to reflect its actual value, and what do calibration and abstention retain when visual evidence or the source domain changes?*

Our contributions are: (1) an executed comparison of batch and fixed-threshold switching that separates volume from selected correction quality (Fig. 3); (2) a sample-level explanation of a sparse calibration plateau and an exhaustive actor-subset sensitivity analysis (Figs. 4,6); (3) a locked external finding that support rejection can collapse clean coverage, with paired source-level uncertainty (Fig. 5, Tab. 2); and (4) new external head inference across three fusion families and five checkpoints, alongside preprocessing and equal-volume controls that separate admission volume from ranking failure. These contributions concern the tested procedures and populations.

2. Problem formulation

Let $a(x)$ and $f(x)$ be the audio and audio–visual predictions, y the intended label, and $D = \{a \neq f\}$ the disagreement set. Define the signed correctness change

$$\Delta = \mathbf{1}\{f = y\} - \mathbf{1}\{a = y\} \in \{-1, 0, 1\}. \tag{1}$$

Rescues have $\Delta = 1$; harms have $\Delta = -1$. Neutral disagreements have both experts wrong and must not be dropped. A switching policy $S(x) \in \{0, 1\}$ returns fusion when $S = 1$ and audio otherwise. Its gain over audio is $G = \mathbb{E}[S\Delta]$. We report the switch fraction $V = \mathbb{E}[S]$, rescue fraction $R = \mathbb{E}[S\mathbf{1}\{\Delta = 1\}]$, and harm fraction $H = \mathbb{E}[S\mathbf{1}\{\Delta = -1\}]$, so $G = R - H$.

For $V > 0$, define selected quality $Q = \mathbb{E}[\Delta \mid S = 1]$; then $G = VQ$. This identity distinguishes changing admission volume from changing admitted outcomes. It is accounting, not an architectural contribution. An audio-versus-fusion oracle gains $\Pr(\Delta = 1)$, but that quantity is not a guarantee that an observable selector can identify rescues.

A fixed policy switches on disagreements with score $s(x) > t$, retaining its clean-tuned threshold under shift. A batch policy sorts disagreements by score and takes a prefix subject to an actor-weighted budget. These answer different deployment questions. Positive-score variants additionally require $s > 0$ and may use less than the budget. Rank-only matched-rate controls intentionally omit positivity to compare ranking at the same actual volume; they are diagnostics, not the proposed deployment rule.

3. Study design and methods

3.1. Experts and observable inputs

Figure 2 describes the implemented pipeline. WavLM Base audio features and DINOv2 ViT-B/14 visual features are frozen. Eight uniformly spaced frames are decoded, letterboxed to 224×224, encoded, and averaged. Each modality feature is projected to 96 dimensions with layer normalization and GELU. The main fusion head applies four-head attention to the two projected modality tokens with a residual connection; their concatenation feeds a six-class MLP. Audio and video controls zero the other projected token. Expert training uses actor-weighted cross entropy and selects a checkpoint on clean development actors. The selectors operate after these heads, so no encoder saving is attributed to routing.

The score representation contains fifteen statistics: maximum probability, top-two margin, and normalized entropy for each expert, plus three pairwise Jensen–Shannon divergences and three prediction-agreement indicators. The original pipeline centers and scales each expert's six logits within a clip before softmax. This removes expert-specific offset and positive scale, stabilizing comparisons between heads, but also discards logit-scale confidence information. Section 5.7 tests that choice instead of assuming invariance is beneficial.

Eight additional visual measurements quantify temporal and crop cosine distances (mean and maximum), central-patch intensity standard deviation, spatial gradient, temporal change, and mean intensity. Crop comparisons require four extra encodings of resized central crops. These are observable statistics, not semantic visibility labels. A constant occluder can suppress temporal variation and appear stable. No reference clean frame is supplied to the predictor at in-

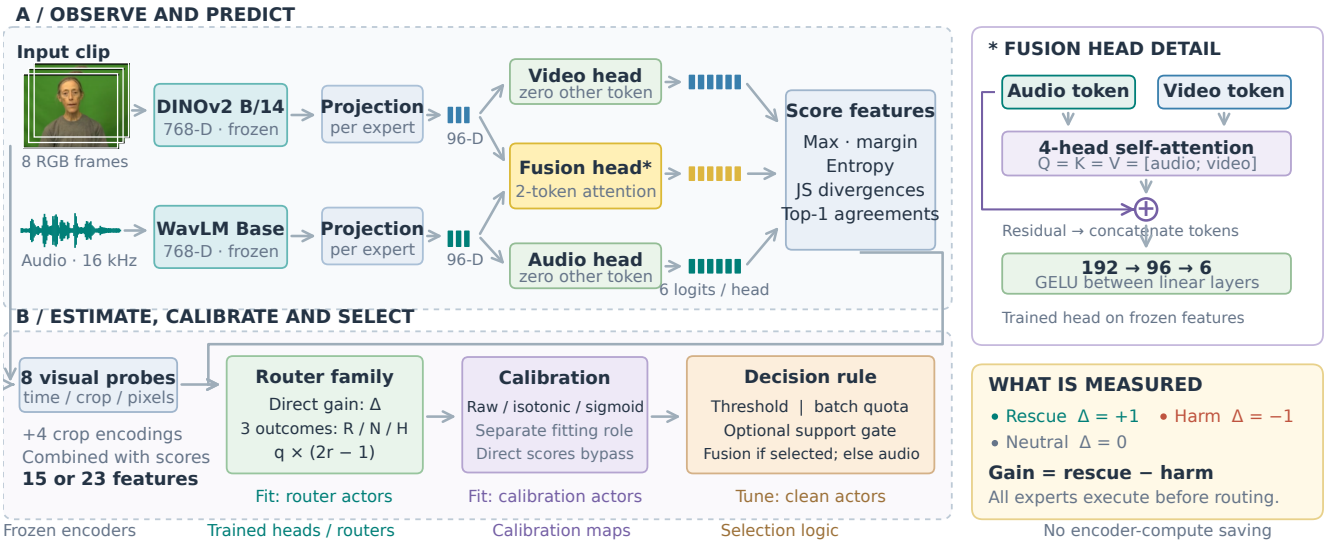


Figure 2. **Executed architecture and decision pipeline.** (A) Frozen encoders feed expert-specific 768-to-96 projections (layer normalization and GELU) and six-class heads; fan-out denotes common input features, not shared projection weights. The inset details the fusion head's two-token attention, residual connection and MLP. (B) Score statistics and eight visual probes feed routing controls, with separate predictor, calibration and threshold-fitting actors. Crops cost four additional visual encodings. The practical support gate and the separate fixed-region certificate are described in the text. All expert inference precedes selection.

ference.

3.2. Common-information predictors

Direct gain regression minimizes squared error on Δ over router-role disagreements. We compare eight audio/fusion statistics, all fifteen score statistics, and those fifteen plus eight visual probes. A three-class control predicts rescue, neutral, and harm with cross entropy and scores rescue probability minus harm probability. A structured control fits $q = \Pr(\Delta \neq 0 \mid Z, D)$ and $r = \Pr(\Delta = 1 \mid Z, D, \Delta \neq 0)$, then scores $q(2r - 1)$. Its second head excludes neutral cases; the first retains them. This factorization is elementary and is not presented as a novel architecture.

All new tree controls use histogram gradient boosting with 100 iterations, seven leaves, minimum leaf size 20, learning rate .05, and L2 penalty one. For actor g with n_g observations in a role, base weights are proportional to $1/n_g$. After restricting to disagreements or non-neutral disagreements, these original weights are renormalized globally; actors are *not* rebalanced within the conditional subset. This distinction is preserved in the calibration audit and checked against archived outputs.

3.3. Calibration, support, and decisions

Raw heads, isotonic calibration, and sigmoid calibration of clipped probability logits are compared on the same actor roles. Sigmoid fitting uses logistic regression with $C = 1$ and probability clipping to $[10^{-6}, 1 - 10^{-6}]$. Classwise isotonic maps for the three-outcome control are renormalized before computing signed gain. A clean tuning split sets each nonnegative fixed threshold to meet an at-most nominal switch budget, with ties retained as whole score levels. The measured external volume need not match that budget.

For visual probe vector h, the empirical support statistic is $d(h) = \sum_j ((h_j - \mu_j)/\sigma_j)^2$, using clean router moments and a standard-deviation floor of 10^{-5}. Its practical rejection cutoff is the 95th percentile of clean calibration distances, 29.981. Outside this region the policy keeps audio; inside it applies direct gain with visual probes. Distance from a reference distribution is not probability of usable evidence. The experiment asks whether this heuristic preserves useful clean opportunities while avoiding harm.

The formal certificate is separate. Its cutoff and score-region boundary are fixed using router actors only; calibration actors then evaluate bounded mean signed gain within two disjoint regions. The earlier calibration-derived-cutoff argument did not justify fixed-region concentration and is replaced by this construction. With independent actor sampling from the deployment population, a union bound gives simultaneous lower endpoints $\bar{Y}_j - \sqrt{2\log(2/\alpha)/n}$. Only the union of positive-bound regions is certified; further filtering inside those regions is not covered. Neither repaired region qualifies at $n = 9$. The proof and assumptions appear in the supplement.

4. Experimental setup

The six labels are angry, disgust, fearful, happy, neutral, and sad. CREMA-D's original test actors were inspected before

Role	Actors	Clips	Status
CREMA router	9	216	Fit predictors
CREMA calibration	9	216	Fit maps
CREMA tuning	9	216	Select thresholds
CREMA test	10	820	Inspected diagnosis
RAVDESS	24	576	Locked external test

Table 1. **Data roles.** The first three roles are source-separated and each has clean, 25% occluded, and blur-2 observations, totaling 1,944 fitting-side records. Expert fitting and development use earlier disjoint CREMA roles. Later sensitivity analyses reuse the test populations and are exploratory. All targets are six shared intended-emotion labels.

this analysis. RAVDESS selection, policy families, fitted thresholds, and primary comparison were locally hashed before feature inference. We select four full audio–video clips per actor and shared class by predetermined filename hash. Calm and surprise are excluded because the frozen heads have no corresponding outputs. This is six-class transfer, not performance on the full eight-class benchmark.

The main stresses are a static central gray square occupying approximately 50% of the letterboxed frame and a moving striped occluder occupying approximately 35%. They are applied to decoded pixels before encoding, with audio unchanged. Training-side conditions use clean input, a smaller central 25% square, and Gaussian blur with standard deviation two. These interventions test controlled sensitivity, not natural visibility. Figure 1 labels the construction explicitly.

Checkpoint 11 defines the primary external comparison: support-aware fixed routing versus margin at a nominal 20% clean tuning budget. The paired 95% actor interval must be above zero for the local superiority criterion. Checkpoint 22 measures sensitivity under the same encoders. Accuracy, gain, switch, rescue, and harm are actor-uniform. Primary intervals use 5,000 paired actor bootstrap draws. Fixed masks remain fixed; batch analyses must re-rank each resampled actor batch. Additional fixed-policy intervals use 4,000 draws and report the actor count. All intervals are conditional on fitted models and pointwise unless stated otherwise.

The calibration-subset, preprocessing, and additional fusion-head studies were designed after external outcomes were inspected. Their protocols prevent evaluation labels from fitting predictors or thresholds, but this does not restore confirmation status. We retain all tested conditions and checkpoints. Media hashes verify file identity, not absence of pretraining overlap.

5. Results and mechanism analysis

5.1. Does the batch rule conceal deployment failure?

Figure 3 compares identical legacy scores under two rules, not different predictors. On clean CREMA input, the batch policy changes 20.0% of clips and gains 5.98 points; the fixed threshold changes 21.46% and gains 5.85. Under central occlusion, their gains become -7.32 and -22.93 points. The latter changes 53.41% of clips, with 6.83% rescues and 29.76% harms as fractions of all clips. Thus a nominal 20% clean tuning budget does not describe test-time admissions.

The mean score increases by 0.119 under this intervention, with an exploratory paired actor interval $[0.018, 0.223]$, while selected quality becomes negative. Both volume and composition matter: a quota contains the number of harmful switches but does not make its ranking robust. This comparison diagnoses one inspected selector; it does not establish that all confidence scores invert under all perturbations.

5.2. Are predicted-positive switches still beneficial?

Requiring positive estimated gain is a natural safeguard. Here it changes neither the legacy 20% batch result nor the independent-threshold result: selected harmful switches already have positive estimates. The fixed threshold itself is positive. An improvement from limiting volume must therefore not be described as a repair of confidently wrong scores. In the controlled training family, raw, calibrated, and direct signed scores are evaluated with the same positivity rule and their realized coverage is reported. Positive-score selection is a decision constraint; its empirical benefit still depends on score validity under shift.

5.3. What does sparse calibration support explain?

The legacy score factors opportunity and conditional rescue. Thirteen inspected test cases receive near-zero estimated rescue probability, yet eight are rescues. Twelve have exactly zero calibrated conditional rescue; none has a zero opportunity estimate. Their raw values lie within the fitted isotonic input range, so simple out-of-range clipping does not explain the result. Figure 4 locates the empirical zero plateau and its six non-neutral calibration cases from four actors, all harms.

The conditional map estimates rescue only after excluding neutral disagreements. That reduces effective support relative to the already small nine-actor calibration role. Its exact zero describes that fitted sample, not a population impossibility. Sigmoid maps avoid such flat empirical zeros but can still assign unhelpful operating scores. We therefore compare outcomes, not only the appearance of the reliability curve. This evidence diagnoses the recorded plateau; general calibration claims require the broader sensitivity analyses below.

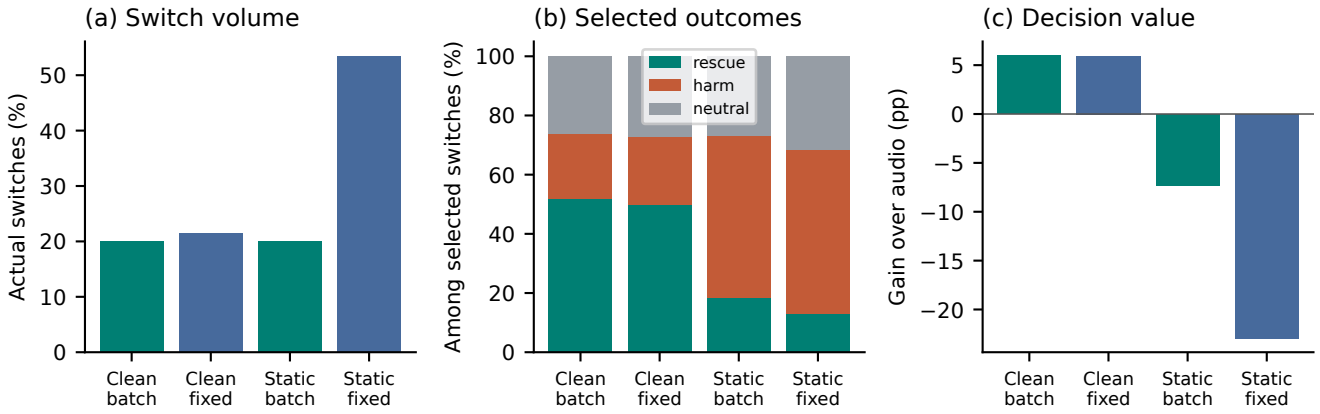


Figure 3. **Batch evaluation understates fixed-threshold deterioration.** On the same 820 inspected CREMA clips from ten actors, (a) the quota constrains switch volume whereas the locked threshold admits 53.4% under constructed occlusion; (b) harm dominates selected outcomes; (c) fixed-threshold loss reaches 22.93 points versus 7.32 under the quota. Bars are actor-uniform point estimates. Paired uncertainty for the score shift is given in the text; these bars are not independent replications.

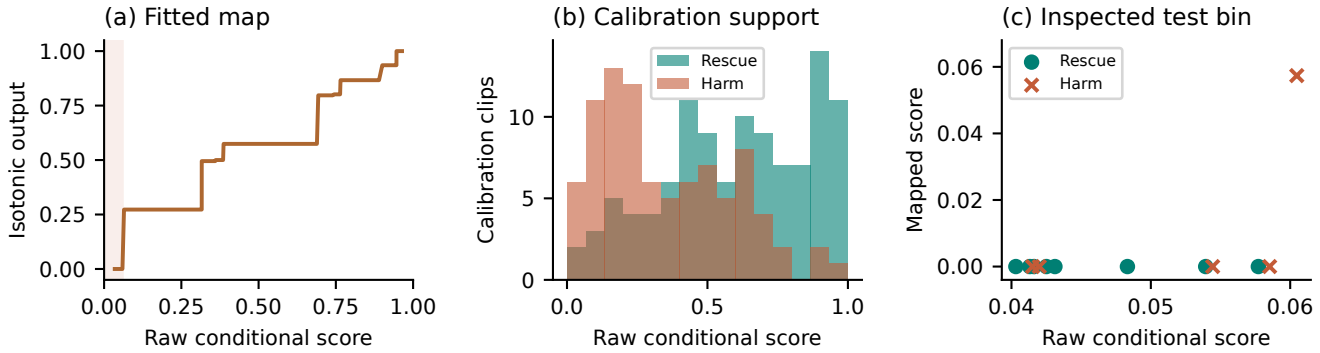


Figure 4. **A fitted zero is not certainty.** (a) The legacy conditional-rescue isotonic map has a zero plateau. (b) Its relevant plateau support consists of six non-neutral calibration clips from four actors, all harms. (c) The inspected near-zero rescue bin contains thirteen clips from six test actors, including eight rescues; twelve map to the plateau rather than falling outside the fitted input range. Fit and inspected-test observations are different sets. This explains one sparse-support failure without claiming isotonic regression generally fails.

5.4. Do visual probes help beyond final scores?

Table 2 foregrounds endpoints and controls. At the locked clean external point, the fifteen-score direct learner reaches 37.50% accuracy, whereas the twenty-three-feature learner reaches 34.72%. Adding the measured temporal, crop, and image statistics does not improve this comparison. The direct and structured twenty-three-feature models also show that factorization alone does not establish an advantage. These are fixed clean-tuned rules with different realized switching, so the table does not attribute a pure ranking effect to architecture.

5.5. Does support rejection retain clean external coverage?

The support-aware rule rejects every RAVDESS clip, including clean input. Its accuracy is consequently audio's 34.38%, versus 38.54% for margin. The paired difference is -4.17 points, with interval $[-6.42, -1.91]$. The failed primary comparison is not replaced by a favorable budget or secondary condition. Figure 5 makes the coverage collapse visible: the reference-distance cutoff treats benign external input as unsupported.

Zero switch-induced harm is not useful robustness when all opportunities are discarded. Fixed fusion reaches 42.71% on this clean external subset, so useful multimodal evidence exists in aggregate despite support rejection. This unrestricted endpoint has a larger switch rate and is not a matched-budget superiority result. The practical implication is to report the cost of abstention alongside harms avoided. The second checkpoint and both constructed external occluders also receive zero support acceptance; these repeated failures share the same reference features and target population.

Policy	Acc.	Gain	Switch	Rescue	Harm	Difference from margin [95% CI]
Audio only	34.38	0.00	0.00	0.00	0.00	-4.17 [-6.42, -1.91]
Fixed fusion	42.71	8.33	37.85	14.93	6.60	+4.17 [+0.17, +7.81]
Margin	38.54	4.17	14.06	5.56	1.39	+0.00 [+0.00, +0.00]
Direct, 15 scores	37.50	3.12	9.03	3.47	0.35	-1.04 [-2.78, +0.69]
Direct, 23 features	34.72	0.35	2.26	0.35	0.00	-3.82 [-6.08, -1.56]
Three outcomes, 23	35.24	0.87	3.82	1.04	0.17	-3.30 [-5.21, -1.22]
Structured raw, 23	36.11	1.74	5.73	1.91	0.17	-2.43 [-4.34, -0.17]
Structured isotonic, 23	35.24	0.87	3.82	1.04	0.17	-3.30 [-5.38, -1.22]
Structured sigmoid, 23	35.24	0.87	3.12	0.87	0.00	-3.30 [-5.38, -1.04]
Support gate, 23	34.38	0.00	0.00	0.00	0.00	-4.17 [-6.42, -1.91]

Table 2. **Locked clean RAVDESS comparison, checkpoint 11, 576 clips and 24 actors.** Accuracy, gain, switch, rescue and harm are percentages or percentage points. The final column is a paired actor difference from margin, with a 95% percentile interval. Audio and fixed fusion are endpoints without the 20% tuning constraint; other rows use frozen thresholds tuned at that nominal budget. Actual switching is explicit. No row is bolded as a method victory.

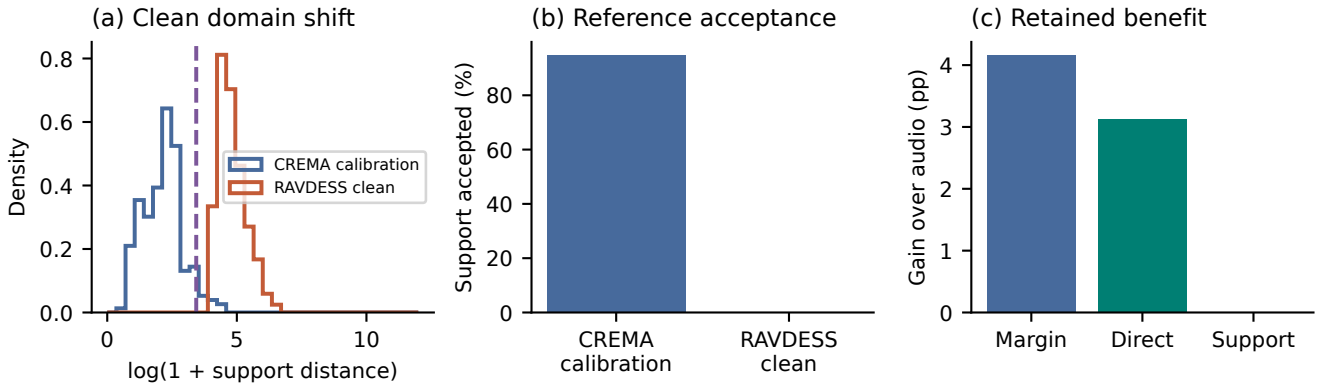


Figure 5. **Support is domain proximity, not evidence quality.** (a) Clean CREMA calibration distances (216 clips, nine actors) and clean RAVDESS distances (576 clips, 24 actors), with the practical cutoff at 29.981. (b) No external clean input is accepted. (c) Support rejection therefore retains none of the gain obtained by margin or score-only direct routing. The separate certificate uses a router-only cutoff and is not the empirical policy plotted here.

5.6. Does lower score error imply more routing utility?

The exhaustive audit freezes the trained heads and refits maps on each actor subset. Clean tuning actors select non-negative thresholds separately for each fitted map. There are 169 subsets per checkpoint, three score variants, and five archived conditions, yielding 5,070 metric rows. Overlapping subsets are sensitivity measurements, not thousands of independent experiments. Full-nine-actor settings reproduce thirty archived policy cases under the original weighting.

With checkpoint 11 and all nine actors, sigmoid calibration lowers clean external signed-score MSE from 1.140 to 0.891, while gain falls from 1.74 to 0.87 points and switching from 5.73% to 3.12%. Isotonic gives MSE 0.907 and the same 0.87-point gain. Checkpoint 22 exhibits a smaller sigmoid gain decline, 5.73 to 5.56 points, despite reducing MSE from 0.795 to 0.734. A population-average score loss weights all disagreements; routing depends on the selected score region. Their improvements need not agree, particularly when coverage changes.

Three-actor isotonic gains span 0.00–3.12 points for checkpoint 11 and 0.17–8.51 for checkpoint 22. Reporting only the best subset would conceal this dependence. Full outputs include rescue/harm Brier scores, signed bias inside the selected region, and exact-zero fractions. These results motivate decision-region evaluation, not a universal preference for an uncalibrated head.

5.7. Does preprocessing remove useful information?

We refit the same direct learner using standardized logits, raw logits, or temperature-scaled logits. One temperature per expert minimizes actor-weighted cross entropy on the calibration role. This uses extra label information compared with the raw and standardized variants. Router fitting and clean threshold tuning retain their original roles; external

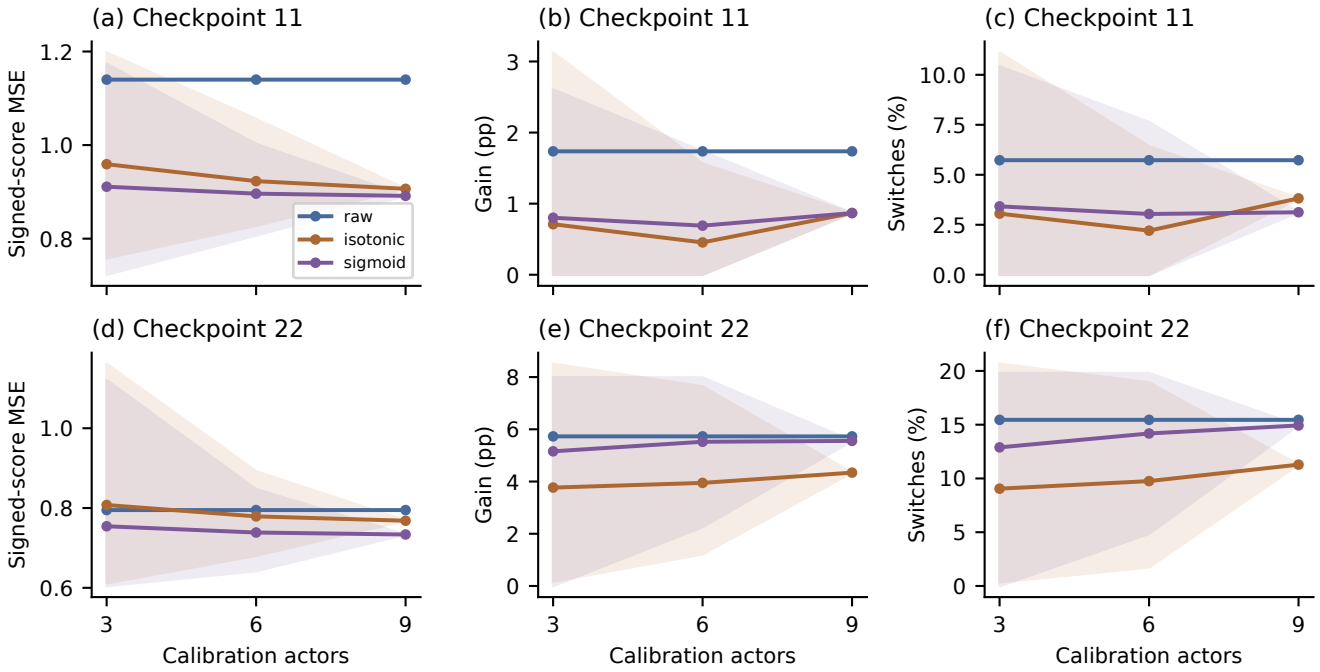


Figure 6. **Calibration composition changes decisions.** Each checkpoint enumerates all 84 three-actor subsets, 84 six-actor subsets, and the full nine calibration actors. Columns show signed-score MSE on disagreements, gain, and switching on clean RAVDESS. Lines average subsets; shading spans their minimum and maximum, not a confidence interval. The raw reference is unchanged across subsets. The 24 evaluation actors are reused, and the two checkpoints share encoders.



Figure 7. **Qualitative cases include successes and failures.** Each column is the lexicographically first archived-gallery example in its stated category; every displayed method uses that same input. (a,b) CREMA clean examples under the legacy policy; (c) clean external missed rescue; (d) clean external rejection. All displayed frames are clean. Predictions use whole clips. Intended labels, not private emotional states, define correctness. These examples illustrate measured decisions and do not estimate prevalence.

labels are used only for evaluation. Ten standardized refits reproduce the archived scores within 10^{-10}.

Table 3 reports fixed and equal-volume clean external decisions. Raw and temperature scores outperform standardized scores at the matched rate in both checkpoints. Fixed-threshold results are less uniform because their coverage differs. Thus removing logit scale can remove useful ranking information; standardization is not a harmless implementation detail. This is an inspected-data sensitivity result, not a selected new method. All five archived conditions and 5%,

Seed	Softmax input	Fixed G	Fixed V	Matched G
11	Standardized	3.12	9.03	3.99
11	Raw	2.60	7.29	5.56
11	Temperature	2.26	5.56	6.25
22	Standardized	5.90	11.98	7.64
22	Raw	9.55	19.79	9.72
22	Temperature	8.85	18.40	8.68

Table 3. Exploratory preprocessing control on clean RAVDESS. G: gain (points); V: switching (%). Matched ranking switches on 19.97% for every row; fixed rules use a clean-tuned nominal 20% budget.

10%, and 20% rank-only budgets are released.

5.8. How dependent is the failure on the fusion head?

We execute new external inference with three existing trained fusion-head families and five checkpoints per family: concatenation, four-head attention, and a residual control matched to the attention head's parameter count. Audio/video experts are checkpoint-matched. Each pair fits fifteen-statistic direct and structured selectors on the original roles; a margin gate is included. The resulting 270 policy comparisons cover clean, static, and moving-occluder input under fixed and matched-rate rules.

Static-occluder direct-routing gain is negative in all fifteen head/checkpoint combinations; margin is negative in all fifteen and calibrated structured routing in fourteen, with one zero. Clean gains remain positive for all direct variants (Tab. 4). Yet the mean direct score rises in only six of fifteen static-occluder comparisons. Harmful switching therefore recurs more broadly than the specific score-increase mechanism. The moving occluder also reduces mean gain in all three head families. At a matched 19.97% switch rate, static-occluder direct gain is still negative in fourteen of fifteen combinations (mean -2.05 points; the remaining gain is $+0.17$). Excess admission volume alone therefore cannot explain the losses: ranking at a common volume also fails. Clean matched gains remain positive in all fifteen. These controls broaden the fusion-head evidence, not the encoder or dataset coverage. Fixed-policy actor intervals and every individual outcome are in the supplement.

6. Related work

Audio–visual datasets and benchmarks expose complementarity and modality dependence [1–3, 38]. Representation and fusion methods include contrastive audio–visual learning and adaptive weighting [4–9, 15–17, 24]. A larger oracle gap does not by itself show that a post-fusion selector can recover its useful cases [2, 3]. Our analysis isolates the deployment policy and its correction outcomes, rather than

Fusion head	Clean	Static	Moving
Late concatenation	+6.70	-2.40	-2.01
Attention	+4.37	-3.37	-2.99
Residual control	+6.15	-2.57	-1.60

Table 4. Exploratory head control: mean fixed-threshold direct-routing gain over five checkpoints, in points. All share frozen encoders and the same 576 external clips. These are repeated-model measurements, not independent populations.

claiming a new representation or inferring architectural novelty from a score factorization.

Conditional computation, expert routing and multimodal uncertainty methods [23, 28–30, 33] motivate adaptive decisions, but post-fusion selection has already paid the expert inference cost. Calibration work distinguishes probability quality from raw confidence [10, 12–14, 32, 41]; our actor-subset audit examines a particular operational consequence under inspected shift. Selective prediction and abstention [11, 31, 35] require useful retained coverage, not merely zero empirical harm. Out-of-distribution detection and broader shift benchmarks [34, 37] contextualize why reference support may reject clean unfamiliar data.

Conformal and risk-control methods state assumptions about calibration and target sampling [20–22, 36]. Our repaired finite-region bound is deliberately narrower and certifies no useful action at the available source count. Natural active-speaker and evidence-association tasks [39, 40] require different temporal data and baselines; the separate object-memory pilot supplies no evidence for this emotion-routing study.

7. Discussion and conclusion

The experiments support an operational conclusion: evaluate the rule that will actually deploy, then report switch volume, selected rescue and harm, and the clean opportunities lost to abstention. A bounded batch can conceal a larger fixed-threshold loss; a fitted calibration zero can rest on little conditional support; and domain rejection can eliminate every external clean opportunity. Neither aggregate calibration error nor zero empirical harm is sufficient to characterize decision value.

The scope remains constrained. The sources use acted intended labels, the occlusions are constructed, the encoders are shared across head controls, and later analyses reuse inspected data. Raw-input reproduction requires official media and encoder weights; portable replay uses logits and probes. The repaired certificate assumes independent actors from the deployment population and gives no arbitrary-shift guarantee. A broader claim needs independent encoder families and natural visibility variation; a redesigned winning method needs a new untouched evaluation.

References

[1] Zverev, Daniil, Wiedemer, Thaddäus, Prabhu, Ameya et al.. VGGSounder: Audio-Visual Evaluations for Foundation Models. 2025. https://arxiv.org/abs/2508.08237.

[2] Hurt, Benjamin. Oracle Complementarity Is Not Realizable Complementarity in Frozen-Encoder Audio-Visual Emotion Recognition. 2026. https://arxiv.org/abs/2609.31764.

[3] Shihab, Ibne Farabi, Ahsan, Abu Sa-Adat Mohamed Moon-Im Al, Swaqeeb, Md Najmus. Opportunity Is Not Realizability: Selection-Valid Diagnostics for Multi-LLM Routing. 2026. https://arxiv.org/abs/2608.08265.

[4] Nielsen, Frank. Generalized Bhattacharyya and Chernoff upper bounds on Bayes error using quasi-arithmetic means. 2014. https://arxiv.org/abs/1401.4788.

[5] Lin, Yan-Bo, Bertasius, Gedas. Siamese Vision Transformers are Scalable Audio-visual Learners. 2024. https://arxiv.org/abs/2403.19638.

[6] Kim, Jongsuk, Lee, Hyeongkeun, Rho, Kyeongha et al.. EquiAV: Leveraging Equivariance for Audio-Visual Contrastive Learning. 2024. https://arxiv.org/abs/2403.09502.

[7] Gong, Yuan, Rouditchenko, Andrew, Liu, Alexander H. et al.. Contrastive Audio-Visual Masked Autoencoder. 2022. https://arxiv.org/abs/2210.07839.

[8] Chen, Honglie, Xie, Weidi, Vedaldi, Andrea et al.. VGGSound: A Large-scale Audio-Visual Dataset. 2020. https://arxiv.org/abs/2004.14368.

[9] Guo, Chuan, Pleiss, Geoff, Sun, Yu et al.. On Calibration of Modern Neural Networks. 2017. https://arxiv.org/abs/1706.04599.

[10] Liang, Shiyu, Li, Yixuan, Srikant, R.. Enhancing The Reliability of Out-of-distribution Image Detection in Neural Networks. 2017. https://arxiv.org/abs/1706.02690.

[11] Geifman, Yonatan, El-Yaniv, Ran. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. 2019. https://arxiv.org/abs/1901.09192.

[12] Gal, Yarin, Ghahramani, Zoubin. Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. 2015. https://arxiv.org/abs/1506.02142.

[13] Lakshminarayanan, Balaji, Pritzel, Alexander, Blundell, Charles. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. 2016. https://arxiv.org/abs/1612.01474.

[14] Kendall, Alex, Gal, Yarin. What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?. 2017. https://arxiv.org/abs/1703.04977.

[15] Wang, Weiyao, Tran, Du, Feiszli, Matt. What Makes Training Multi-Modal Classification Networks Hard?. 2019. https://arxiv.org/abs/1905.12681.

[16] Peng, Xiaokang, Wei, Yake, Deng, Andong et al.. Balanced Multimodal Learning via On-the-fly Gradient Modulation. 2022. https://arxiv.org/abs/2203.15332.

[17] Nagrani, Arsha, Yang, Shan, Arnab, Anurag et al.. Attention Bottlenecks for Multimodal Fusion. 2021. https://arxiv.org/abs/2107.00135.

[18] Vaswani, Ashish, Shazeer, Noam, Parmar, Niki et al.. Attention Is All You Need. 2017. https://arxiv.org/abs/1706.03762.

[19] Dosovitskiy, Alexey, Beyer, Lucas, Kolesnikov, Alexander et al.. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. 2020. https://arxiv.org/abs/2010.11929.

[20] Angelopoulos, Anastasios N., Bates, Stephen. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. 2021. https://arxiv.org/abs/2107.07511.

[21] Angelopoulos, Anastasios N., Bates, Stephen, Fisch, Adam et al.. Conformal Risk Control. 2022. https://arxiv.org/abs/2208.02814.

[22] Angelopoulos, Anastasios N., Bates, Stephen, Candès, Emmanuel J. et al.. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. 2021. https://arxiv.org/abs/2110.01052.

[23] Nixon, Jeremy, Dusenberry, Mike, Jerfel, Ghassen et al.. Measuring Calibration in Deep Learning. 2019. https://arxiv.org/abs/1904.01685.

[24] Aytar, Yusuf, Vondrick, Carl, Torralba, Antonio. SoundNet: Learning Sound Representations from Unlabeled Video. 2016. https://arxiv.org/abs/1610.09001.

[25] Sanyuan Chen et al.. WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing. 2021.. https://arxiv.org/abs/2110.13900.

[26] Maxime Oquab et al.. DINOv2: Learning Robust Visual Features without Supervision. 2023.. https://arxiv.org/abs/2304.07193.

[27] Houwei Cao, David G. Cooper, Michael K. Keutmann et al.. CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset. 2014. https://github.com/CheyneyComputerScience/CREMA-D.

[28] Hong, Joanna, Kim, Minsu, Choi, Jeongsoo et al.. Watch or Listen: Robust Audio-Visual Speech Recognition with Visual Corruption Modeling and Reliability Scoring. 2023. https://arxiv.org/abs/2303.08536.

[29] Zhang, Qingyang, Wei, Yake, Han, Zongbo et al.. Multimodal Fusion on Low-quality Data: A Comprehensive Survey. 2024. https://arxiv.org/abs/2404.18947.

[30] Ovadia, Yaniv, Fertig, Emily, Ren, Jie et al.. Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. 2019. https://arxiv.org/abs/1906.02530.

[31] Geifman, Yonatan, El-Yaniv, Ran. Selective Classification for Deep Neural Networks. 2017. https://arxiv.org/abs/1705.08500.

[32] Cattelan, Luís Felipe P., Silva, Danilo. How to Fix a Broken Confidence Estimator: Evaluating Post-hoc Methods for Selective Classification with Deep Neural Networks. 2023. https://arxiv.org/abs/2305.15508.

[33] Kull, Meelis, Perello-Nieto, Miquel, Kängsepp, Markus et al.. Beyond temperature scaling: Obtaining well-calibrated

multiclass probabilities with Dirichlet calibration. 2019. https://arxiv.org/abs/1910.12656.

[34] Hendrycks, Dan, Gimpel, Kevin. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. 2016. https://arxiv.org/abs/1610.02136.

[35] Ziyin, Liu, Wang, Zhikang, Liang, Paul Pu et al.. Deep Gamblers: Learning to Abstain with Portfolio Theory. 2019. https://arxiv.org/abs/1907.00208.

[36] Tibshirani, Ryan J., Barber, Rina Foygel, Candes, Emmanuel J. et al.. Conformal Prediction Under Covariate Shift. 2019. https://arxiv.org/abs/1904.06019.

[37] Koh, Pang Wei, Sagawa, Shiori, Marklund, Henrik et al.. WILDS: A Benchmark of in-the-Wild Distribution Shifts. 2020. https://arxiv.org/abs/2012.07421.

[38] Steven R. Livingstone, Frank A. Russo. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English. 2018. https://doi.org/10.1371/journal.pone.0196391.

[39] Ruijie Tao, Zexu Pan, Rohan Kumar Das, Xinyuan Qian, Mike Zheng Shou, and Haizhou Li. Is Someone Speaking? Exploring Long-term Temporal Features for Audio-visual Active Speaker Detection. ACM Multimedia, 2021. https://arxiv.org/abs/2107.06592.

[40] Yidi Jiang, Ruijie Tao, Zexu Pan, and Haizhou Li. Target Active Speaker Detection with Audio-visual Cues. Interspeech, 2023. https://arxiv.org/abs/2305.12831.

[41] Meelis Kull, Telmo Silva Filho, and Peter Flach. Beta calibration: a well-founded and easily implemented improvement on logistic calibration for binary classifiers. AISTATS, 2017. https://proceedings.mlr.press/v54/kull17a.html.